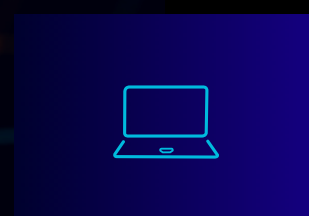




TYPES OF EXECUTIONS



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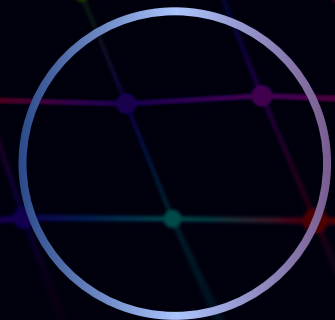
Unidades tecnologicas de Santander



TYPES OF EXECUTIONS

Computer processors only understand machine language (zeros and ones). Since humans write code in high-level languages (text), this code must be translated.

The "type of execution" refers to when and how this translation occurs. There are three main approaches:





COMPILED LANGUAGES

A special program called a compiler reads all your source code and translates it into machine language all at once before the program is executed. This generates a final executable file (like an .exe in Windows).

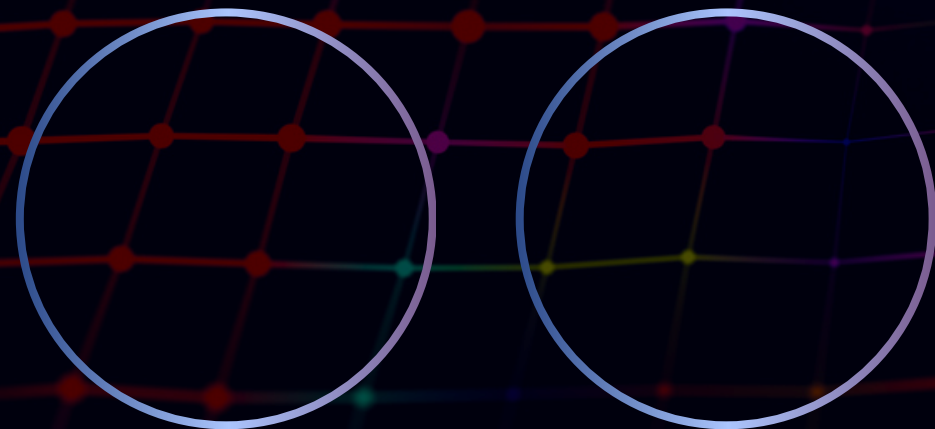


COMPILED LANGUAGES

- The biggest advantage: They are extremely fast and efficient, as the processor executes the machine code directly without intermediaries.
- The biggest drawback: They are platform-dependent. If you compile a program for Windows, that same file won't work on Linux or macOS; you would have to recompile the code specifically for those systems.
- Classic examples: C, C++, Go, Rust.



WHAT TYPES OF PROGRAMS REQUIRE COMPILEATION?

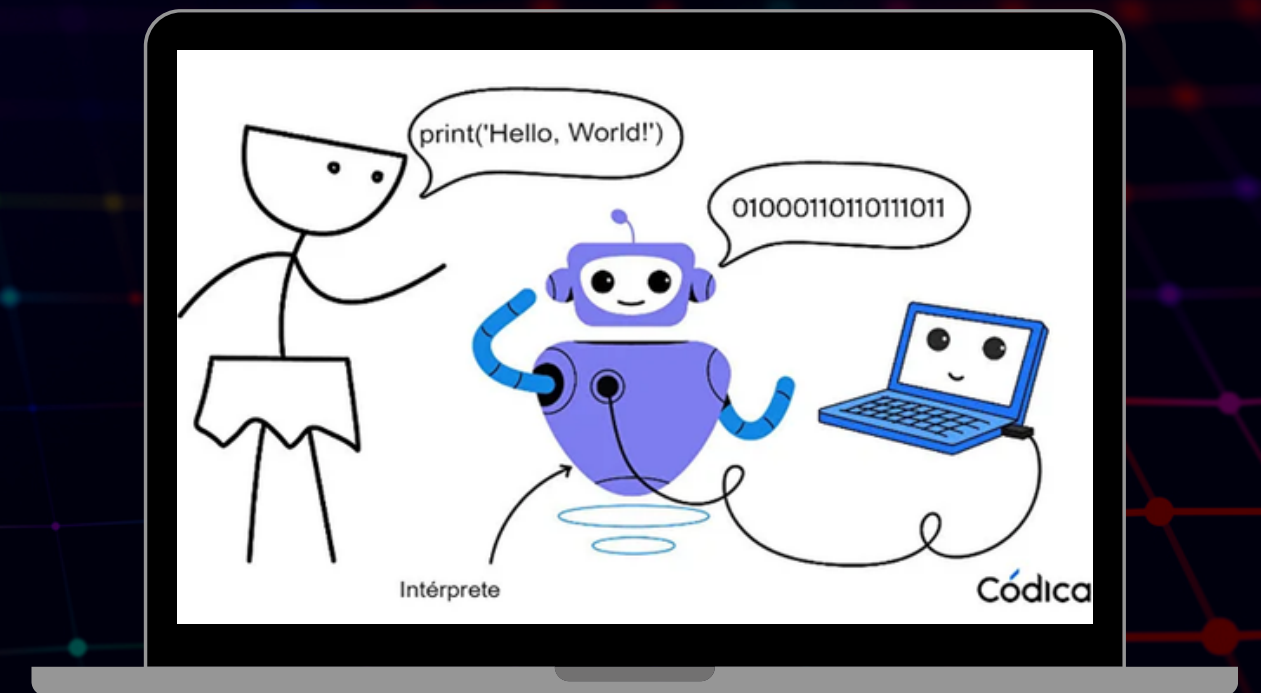


Programs that require extensive calculations, such as game engines, multimedia applications, operating systems, and programming tools (such as IDEs), often require compilation to optimize their performance.



INTERPRETED LANGUAGES

In this model, no prior executable file is created. A program called an interpreter reads your source code and translates it into machine language line by line, executing it in real-time.





INTERPRETED LANGUAGES



- The biggest advantage: They are highly portable. The same code will work on any computer or operating system, as long as it has the corresponding interpreter installed. Additionally, testing and debugging are much more agile.
- The biggest drawback: They are slower than compiled languages, since the translation process happens simultaneously with execution.
- Classic examples: Python, JavaScript, Ruby, PHP.

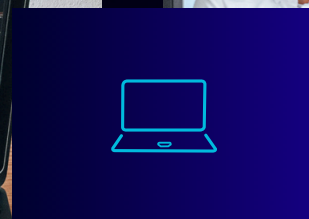
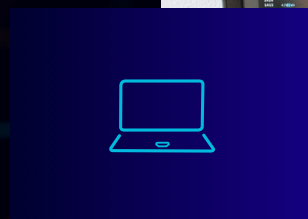


TYPES OF PROGRAMS AND COMMON USES

- Automation and scripts: System or day-to-day tasks.
- Web development (backend and frontend): Server logic and interactive pages.
- Data science and artificial intelligence: Analysis of numbers and advanced models.
- Education and prototypes: Rapid testing of ideas due to their ease of use.



HYBRID LANGUAGES (BYTECODE / VIRTUAL MACHINE)



These seek the best of both worlds. The process happens in two steps: first, the source code is compiled into an intermediate language (known as bytecode), which isn't machine code, but is very close to it. Then, a program called a Virtual Machine reads that bytecode and executes it on your computer.



HYBRID LANGUAGES (BYTECODE / VIRTUAL MACHINE)

- The biggest advantage: Perfect balance. They are almost as fast as compiled languages and as portable as interpreted ones (the famous "Write once, run anywhere" principle).
- The biggest drawback: For the program to work, the end-user must have the specific Virtual Machine installed on their device.
- Classic examples: Java (uses the Java Virtual Machine or JVM), C# (uses the .NET environment), Kotlin.





EXAMPLES OF PROGRAMS AND LANGUAGES WITH BYTECODE

- Java programs: They are compiled into files .class containing bytecode and run on the Java Virtual Machine (JVM)

- C# (.net) apps: : They are transformed into an intermediate code called CIL (Common Intermediate Language) that runs on the CLR (Common Language Runtime).

- Python scripts: The code is internally translated into bytecode files (.pyc) and processed by the [Python Virtual Machine].





REFERENCE



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THANK
YOU!